

1 Boltzmann generators on molecular potential energies

The model problems of the previous section — the 2D benchmarks, the lattice φ^4 theory, the screened Poisson inversion — were each chosen to isolate one failure mode of forward KL training on a smooth target. The application that motivates a Boltzmann generator in the first place is a molecule, and a molecular target stresses the whole pipeline at once. This section applies the adaptive-temperature, X-regularized generator of the previous section to three molecules of increasing difficulty in full internal coordinates, $d = 36$ to 60, on a single consumer GPU. It is organized into four parts: the motivation and the two difficulties that make a molecular potential hard (§1.1); *sharpening*, the single trick that makes the framework tractable on a singular target (§1.2); the molecule-specific engineering it requires (§1.3); and the numerical results on glycerol, diethanolamine, and alanine dipeptide (§1.4).

1.1 Motivation and background

The targets studied so far were smooth by design. The distribution that motivates a Boltzmann generator is a molecule: the equilibrium (Boltzmann) distribution $\mu \propto e^{-U_1}$ of a molecular potential energy U_1 , whose samples are the conformers that set binding affinities, reaction rates, and free energies in computational chemistry and drug design (Noé et al., 2019). Drawing *independent* samples from μ is the central computational problem: molecular dynamics and Markov-chain Monte Carlo mix slowly across the high torsion barriers that separate conformers, so a trained flow that emits independent conformers with a tractable density — and therefore the exact importance-sampling pipeline of the previous sections — is the payoff that justifies the training cost.

We take U_1 to be the Amber molecular-mechanics force field (Cornell et al., 1995), the standard classical model: a sum of bonded terms (harmonic bonds and angles, periodic torsions) and non-bonded Lennard-Jones and Coulomb terms,

$$U_1(x) = \sum_{\text{bonds}} k_b(r-r_0)^2 + \sum_{\text{angles}} k_\theta(\theta-\theta_0)^2 + \sum_{\text{torsions}} \frac{V_n}{2} [1 + \cos(n\phi - \gamma)] + \sum_{i < j} \left[4\epsilon_{ij} \left(\frac{\sigma_{ij}^{12}}{r_{ij}^{12}} - \frac{\sigma_{ij}^6}{r_{ij}^6} \right) + \frac{q_i q_j}{r_{ij}} \right], \quad (1)$$

the last sum running over non-bonded atom pairs (charges in units that absorb the Coulomb constant). We evaluate U_1 for the *isolated molecule in vacuum* (no solvent, no nonbonded cutoff). Two simplifications bound these experiments and are genuine limitations left to future work. (i) *Vacuum*: the sampler is solvent-agnostic — solvation only reshapes U_1 , not the algorithm — and the vacuum targets already display, qualitatively, the feature we want to illustrate (the well-separated, crucial potential wells the generator must discover and resolve), but implicit or explicit solvent and larger peptides are not treated here. (ii) *No chirality preservation*: the internal-coordinate generator carries no handedness constraint, so for the achiral force field it samples *both* the L- and D-enantiomers (§1.4), whereas a physical trajectory stays in one; isolating a single enantiomer by a filter or restraint is left to future work.

We work in *whitened internal coordinates* (Noé et al., 2019). The six rigid-body degrees of freedom (global translation and rotation) leave U_1 unchanged, so we drop them and describe the shape by bond lengths, bond angles, and torsions (a z-matrix) — a smooth, invertible map $x \mapsto \xi \in \mathbb{R}^d$ with $d = 3N - 6$ and a tractable volume Jacobian. Bonds and angles are stiff and near-Gaussian about their equilibria, so each is standardized to unit variance; torsions stay periodic on $[-\pi, \pi)$. The source $\mu_0 \propto e^{-U_0}$ is then a trivial product base — standard normal on the bond/angle directions, uniform on the torsions — and the target, pulled back through the change of variables, is

$$\mu(\xi) \propto \exp(-U_1(x(\xi))) \left| \det(\partial x / \partial \xi) \right|, \quad (2)$$

so U_0 and U_1 as functions of ξ already absorb the change-of-variables log-Jacobian. This is what makes the target tractable for a flow: the rigid-body symmetries and the stiff, strongly correlated bond/angle vibrations are factored out, leaving the combinatorially multimodal torsional landscape — the conformers — as the structure the diffeomorphism must actually transport.

Two features make (1) qualitatively harder than a smooth model target, and they compound. First, the dimension grows with the molecule ($d = 36$ to 60 here), and the conformer landscape is *combinatorially multimodal*: every rotatable torsion is multi-well (gauche $^\pm$ /trans), so the joint over several torsions has dozens of basins the flow must discover. Second, and more consequential, the Lennard-Jones r^{-12} repulsion makes U_1 *singular*: as any pair distance $r_{ij} \rightarrow 0$ the energy diverges, so the target has a hard, near-singular clash boundary that a smooth diffeomorphism cannot represent without enormous Jacobians. Neither difficulty is fatal in isolation — high-dimensional smooth targets and low-dimensional singular ones are both routine — but together they are worse than either alone: the singular clash core sits *inside* the high-dimensional multimodal landscape, and a single flow asked to discover modes *and* resolve the singularity at once trains badly at both. This “ $1 + 1 > 2$ ” coupling, absent from every model problem of the previous section, is the difficulty the rest of this section is built to manage. On alanine dipeptide ($d = 60$) below it is decisive: the clash core, not the torsion profile, becomes the governing constraint.

1.2 Sharpening: the core simplification

The remedy is to never push the singular part through the flow. We split each stage of the temperature ladder into a *flow step* that advances the temperature at a soft regularization, and a *Monte-Carlo sharpening step* that advances the regularization by importance reweighting alone. Two soft-core controls tame the $r \rightarrow 0$ clash: an energy cap e that log-compresses energies above a threshold, and a distance floor r_{floor} that clamps short-range pair distances. Both are annealed by an index $s \in [0, 1]$ from soft ($s = 0$) to sharp ($s = 1$); the schedule may be *geometric* or *arithmetic* (both are used below). Writing $U_{1,s}$ for the target (1) at regularization s , with $U_{1,1} = U_1$ the sharp target and $U_{0,s} = U_0$ the un-regularized source, the bridge potential is

$$U_{t,s} = (1 - t) U_0 + t U_{1,s}, \quad (3)$$

and the whole run is a single anneal from the soft source $U_{0,0}$ to the sharp target $U_{1,1}$, started at $s_0 = 0$ with $s_k = t_k$. At each stage the flow advances the temperature t at the *lagged* regularization, $U_{t_{k-1},s_{k-1}} \rightarrow U_{t_k,s_{k-1}}$, trained under the X-regularized forward KL $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$; the Monte-Carlo step then sharpens the regularization, $U_{t_k,s_{k-1}} \rightarrow U_{t_k,s_k}$, by one importance reweight and a short Langevin rejuvenation. The soft-core controls act *only* on the high-energy part of the potential, so the anneal loosens the clash barrier without perturbing the low-energy basins and, in principle, loses no modes; the diffeomorphism trains stably on a smooth surrogate while the cheap Monte-Carlo step carries the singularity it cannot. The importance step asks nothing the singularity forbids: it never inverts or differentiates U_1 , only reweights samples and rejuvenates them with a short Langevin move inside the basin they already occupy, so the stiff repulsive wall costs a little weight variance rather than the unbounded Jacobian it would impose on a smooth map. This is the asymmetry the split exploits — the flow handles the part that is smooth but high-dimensional, the importance step the part that is low-dimensional but singular.

Each stage yields two effective sample sizes — the flow’s validation ESS and the Monte-Carlo sharpening ESS, each a normalized fraction in $(0, 1]$ — and the Monte-Carlo error-propagation factor through both per-stage reweights,

$$F = \prod_k \frac{1}{\text{ESS}_k^{\text{val}}} \frac{1}{\text{ESS}_k^{\text{sharp}}}, \quad (F = 1 \text{ ideal}), \quad (4)$$

charges the cost of both (F smaller is better; a no-flow run with no sharpening step has $\text{ESS}^{\text{sharp}} \equiv 1$). The factor F is the headline metric for every result below: it is the variance amplification a downstream importance-sampling estimator inherits from the full ladder. The split lets the flow carry the smooth transport and the importance step carry the singular sharpening.

1.3 Engineering for the singularity

The singular targets require three further devices, all inert on the smooth ones (glycerol and diethanolamine ran with none of them). (i) *Particle drop*: each validation and resample ESS discards the worst few percent of particles before the ESS is formed ($\delta_{\text{drop}} = 0.02$ on alanine dipeptide, 0 on the lighter targets), so a handful of clash-singular outliers — finite but enormous under the soft cap — cannot hijack the importance weights. (ii) *Best-ESS gate snapshots*: within a stage the flow is snapshotted at each ESS gate and the stage keeps the *highest*-validation-ESS snapshot rather than the last training step (which overfits the soft cap and sheds ESS); a step-0 identity snapshot is always included as a Monte-Carlo fallback. (iii) *Distance-floor anneal*: for the worst clash core (alanine dipeptide) r_{floor} is annealed alongside the cap, softening the Lennard-Jones wall early and tightening it to the physical value only at $s = 1$, at the cost of a longer ladder. The wide-coverage pool $\hat{\mu}$ is supplied by the δ -reweighted quench and temper introduced earlier ($\delta = 0.1$), unchanged; the sharpening split remains the core mechanism, and these three devices are the supporting engineering the singularity demands.

A final, metric-level choice accompanies them on the singular target: the per-stage ESS that gates and scores each stage is a *smooth* estimator. Before the ESS is formed, the log importance weights are saturated about their median m by a MAD-scaled tanh, $\log w \mapsto m + s \tanh((\log w - m)/s)$ with $s = c \text{median}|\log w - m|$ ($c = 6$ on alanine dipeptide), which bounds the few clash-singular dominators that otherwise collapse the raw weighted ESS (the WSS) — e.g. a raw ESS of 0.4 reads as a smooth ≈ 0.5 . Glycerol and diethanolamine, free of such dominators, use the raw ESS.

1.4 Numerical results

We close three molecules of increasing difficulty in full internal coordinates on a single consumer GPU (RTX 5070 Ti): glycerol ($d = 36$), diethanolamine ($d = 48$), and alanine dipeptide ($d = 60$, the canonical Boltzmann-generator benchmark). Table 1 summarizes the comparison that runs through all three: the X-regularized δ -reweighted loss against a no-flow reference — the same adaptive sharpening scheme with the flow set to the identity at every stage, which selects its own, longer ladder. Because F (4) is a product over stages, the flow-to-no-flow *ratio*, not the absolute F , is the cross-molecule comparison: the trained flow cuts F by 30–40 $\times$ on the two clash-mild, combinatorially multimodal targets, but only $\sim 2\times$ on alanine dipeptide, where the singular clash core — which the Monte-Carlo sharpening already resolves — rather than the multimodality dominates. The per-molecule subsections below give the full ablation, ladder, and reference figures for each. The baselines here — the bare forward KL and the identity-flow annealed SMC — are the natural same-cost comparisons for a forward-KL-trained generator; we do not benchmark against the flow annealed-importance-sampling bootstrap (FAB) (Midgley et al., 2022), whose large per-step batches and replay buffer make a like-for-like comparison under the same compute budget as these forward-KL-type losses prohibitive, so that comparison too is left to future work.

target	d	K_{flow}	F (X-regularized flow)	F (no-flow reference)
glycerol	36	4	14.16	463.02
diethanolamine	48	5	35.90	1455.44
alanine dipeptide	60	11	67.95	129.70

Table 1: Molecular Boltzmann generators in full internal coordinates. K_{flow} is the trained-flow ladder length; the no-flow reference adaptively chooses its own, longer ladder ($K = 10, 10, 12$ respectively). F (4) is the error-propagation factor (smaller is better, best in bold; F includes the Monte-Carlo sharpening reweight), for the X-regularized δ -reweighted loss against the no-flow annealed-SMC reference. Single-run point estimates; since F is a product over stages, the flow-to-no-flow ratio — not the absolute F — is the cross-molecule comparison.

1.4.1 Glycerol ($d = 36$)

We ablate the generator on the full internal-coordinate glycerol target along three independent axes: the loss (forward KL versus the X-regularized $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$); the regularization schedule (*sharpen*: the soft-core cap is geometrically annealed $e: 100 \rightarrow 200$ along the ladder with a Monte-Carlo sharpening step per stage; *raw*: fixed $e = 200$, no anneal or sharpening); and the wide-coverage sampler (plain quench and temper versus its δ -reweighted variant, $\delta = 0.1$, defined only for the X loss). The headline is the error-propagation factor F (4); the *raw* runs take no sharpening step ($\text{ESS}^{\text{sharp}} \equiv 1$).

loss (raw pushforward)	K	wall (min)	F
annealed SMC	10	2	635.56
forward KL	7	215	40.86
$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$	6	162	21.96
$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, δ -rw.	6	158	16.56
loss (sharpening)	K	wall (min)	F
annealed SMC	10	5	463.02
forward KL	4	88	44.00
$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$	4	73	22.51
$\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, δ -rw.	4	73	14.16

Table 2: Glycerol ($d = 36$) ablation, raw pushforward (top) and sharpening (bottom): ladder length K , training wall-clock, and the error-propagation factor F (4) (smaller better; the sharpening rows include the per-stage Monte-Carlo sharpening reweight). All wall-clock times are on a single RTX 5070 Ti. δ -rw. = δ -reweighted quench and temper.

The X regularizer is the dominant factor, roughly halving F at either schedule. The δ -reweighted quench and temper lowers F further in both schedules ($22.5 \rightarrow 14.2$ under sharpen, $22.0 \rightarrow 16.6$ under raw). Once the sharpening reweight is charged to F , the *sharpen* schedule no longer beats the fixed-cap *raw* run on F alone ($22.5 \approx 22.0$); its advantage is feasibility and speed, reaching $t = 1$ in 4 stages / 73 min versus 6–7 stages / 160–215 min. The lowest- F and fastest run is $\text{KL} + \text{X}_\mu + \text{X}_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted, under the sharpen schedule. Turning the trained flow off entirely — the identity map at every stage, i.e. plain adaptive annealed SMC over the same ladder and acceptance standard — gives $F = 463$ (sharpen) and $F = 636$ (raw), each in $K = 10$ stages. The trained flow thus cuts the propagation factor by ~ 30 – $40\times$ while roughly halving the ladder length: this is the quantitative value the transport map adds over annealing alone.

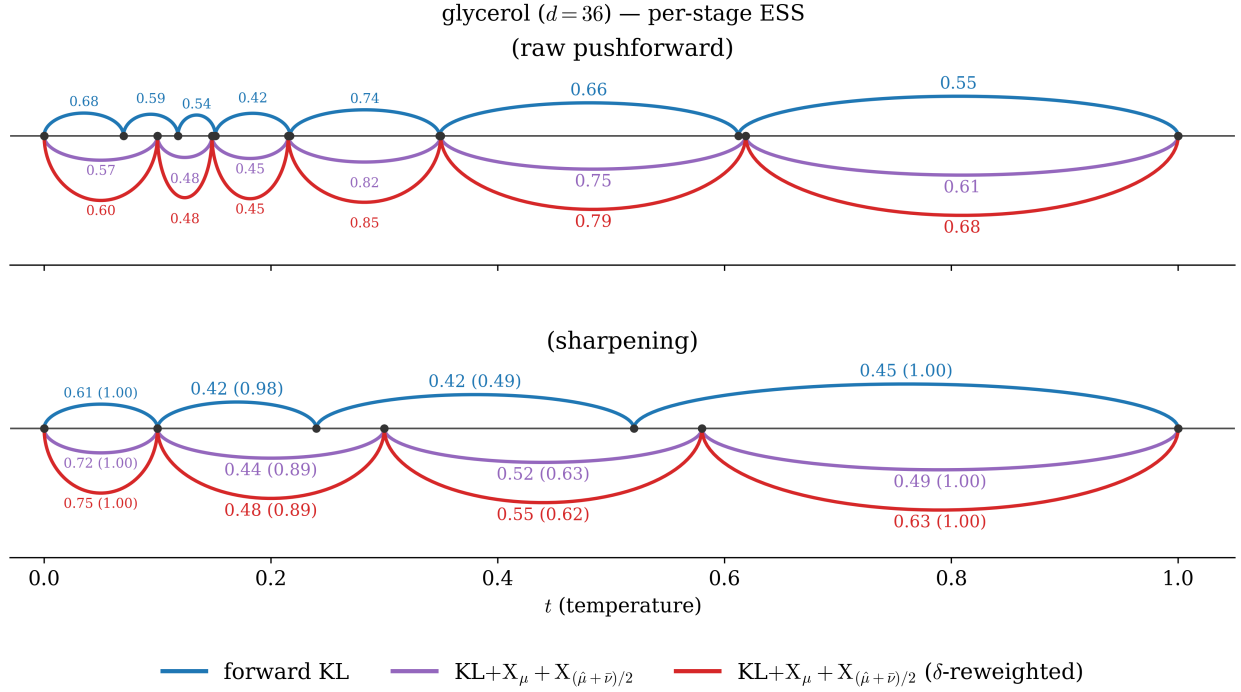


Figure 1: Glycerol per-stage validation ESS as annealing-ladder arcs, raw (top) and sharpen (bottom). In each row forward KL is drawn above the axis (blue), $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ below (purple), and its δ -reweighted variant further below (red). Each arc is an accepted stage; the apex label is that stage’s validation ESS; a cross marks a bridge the ladder failed to reach.

Glycerol is achiral (each torsion marginal is even, $p(\phi) = p(-\phi)$), and every rotatable torsion is three-fold: gauche $^{\pm}$ near $\pm 60^{\circ}$ and trans near 180° . The $\pm\phi$ symmetry pairs gauche $^- \leftrightarrow$ gauche $^+$ and leaves trans self-symmetric, so the joint target over the five rotatable torsions is combinatorially multimodal (dozens of conformers), which is what the wide-coverage pool must cover.

glycerol — backbone O-C-C-O rotamers

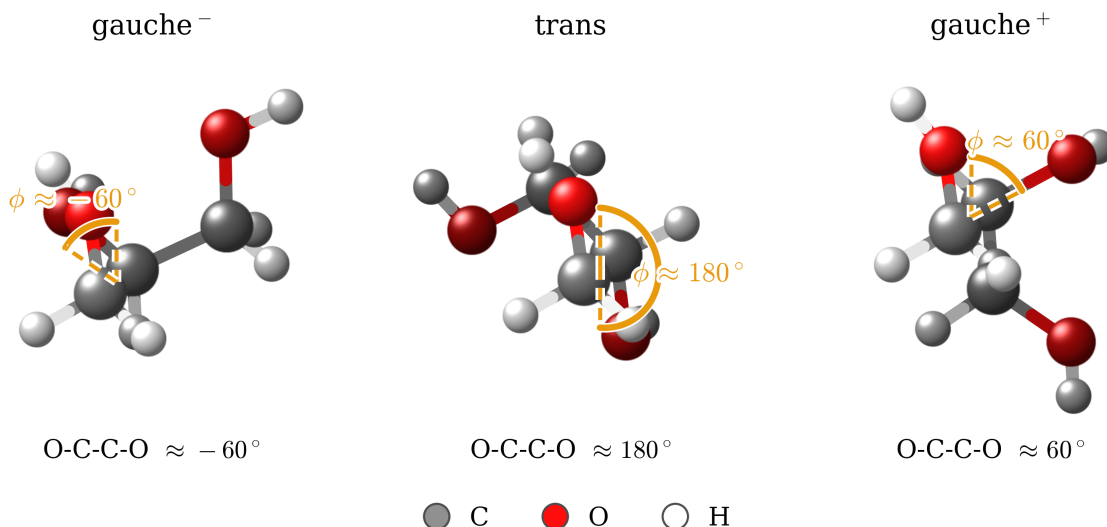


Figure 2: Typical glycerol configurations at the three states of the backbone O-C-C-O torsion: gauche $^-$ (-60°), trans (180°), gauche $^+$ ($+60^{\circ}$). The gold arc marks the dihedral ϕ ; ball-and-stick (C grey, O red, H white). These are the rotamer wells that, taken in combination across the five rotatable torsions, make the target combinatorially multimodal.

glycerol ($d = 36$) — torsion marginals: BG vs annealed SMC reference

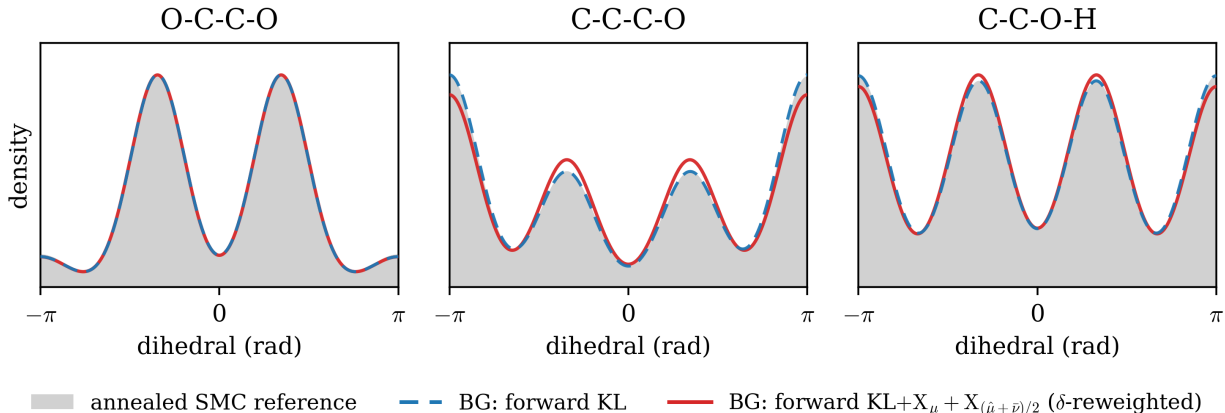


Figure 3: Glycerol torsion marginals of both generators against a long, converged molecular-dynamics reference (grey; a batched Langevin on the same target, 2×10^5 walkers run to convergence): forward KL (blue) and $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -reweighted (red) — both sharpened, the cap annealed to $e = 200$. Because the torsion marginals are insensitive to the soft cap (steric clashes are unpopulated at equilibrium), this reference coincides with the bare-Amber torsion distribution. Both generators recover the multimodal gauche $^\pm$ /trans structure of the characteristic glycerol torsions, evidence that they sample the target rather than collapsing onto a single well.

1.4.2 Diethanolamine ($d = 48$)

We test the generator on the full internal-coordinate diethanolamine target ($\text{HN}(\text{CH}_2\text{CH}_2\text{OH})_2$, $d = 48$), comparing the two best losses under the same sharpening schedule (the soft cap geometrically annealed $e: 100 \rightarrow 200$, a Monte-Carlo sharpening step per stage): forward KL, and the X-regularized $\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$ with the δ -reweighted quench and temper ($\delta = 0.1$). Forward KL reaches $t = 1$ in $K = 6$ stages, the X-regularized loss in $K = 5$ (a single direct $0.65 \rightarrow 1$ step), both on a single RTX 5070 Ti.

loss	K	wall (min)	F
annealed SMC	10	7	1455.44
forward KL	6	896	92.60
$\text{KL} + X_\mu + X_{(\hat{\mu} + \bar{\nu})/2}$, δ -rw.	5	782	35.90

Table 3: Diethanolamine ($d = 48$): two trained-flow losses and a no-flow annealed-SMC reference, all with a per-stage Monte-Carlo sharpening reweight (cap e geometrically annealed $100 \rightarrow 200$): ladder length K , training wall-clock, and the error-propagation factor F (4) (smaller better; F includes the sharpening reweight). All wall-clock times are on a single RTX 5070 Ti. δ -rw. = δ -reweighted quench and temper.

The X-regularized loss with δ -reweighting reaches $F = 35.9$ versus forward KL’s $F = 92.6$ — a $\sim 2.6\times$ reduction in importance-weight waste at identical settings (glycerol showed a comparable $\sim 3.1\times$ gap, 14.2 vs 44.0). The advantage shows in the late, sharp stages and in the ladder length: forward KL’s per-stage validation ESS stalls at 0.46–0.53 and it needs 6 rungs to reach $t = 1$,

whereas the $X+\delta$ run holds a higher ESS (0.74 on the final, direct $0.65 \rightarrow 1$ step) and reaches $t = 1$ in 5 (Fig. 4). This is a singular, high-dimensional, combinatorially multimodal target, so each run is a multi-hour computation; that the X -regularized loss closes it on a single consumer GPU — not a datacenter accelerator — is the practical point.

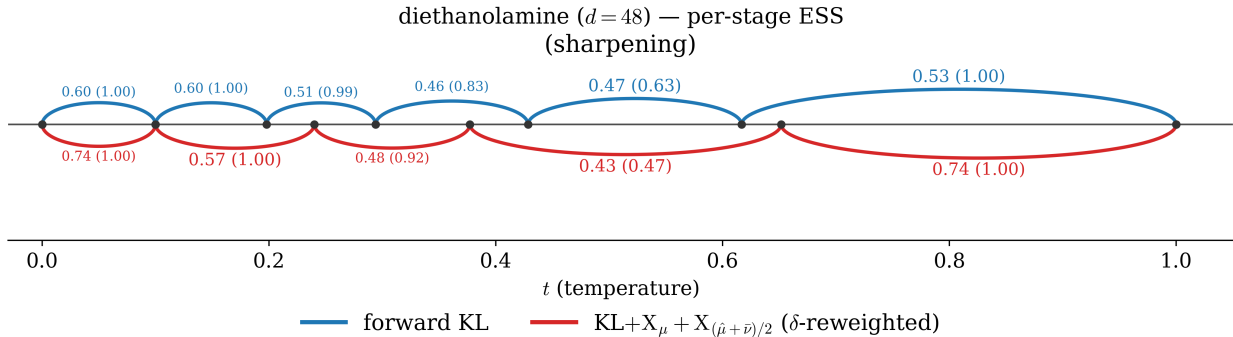


Figure 4: Diethanolamine per-stage ESS as annealing-ladder arcs (sharpening schedule): forward KL above the axis (blue), $\text{KL} + X_\mu + X_{(\hat{\mu} + \hat{\nu})/2}$, δ -reweighted, below (red). Each arc is an accepted stage; the apex label is that stage’s validation ESS, with the sharpening ESS in parentheses.

Diethanolamine has a plane of symmetry (each torsion marginal is even, $p(\phi) = p(-\phi)$); its rotatable backbone torsions — O–C–C–N and C–C–N–C along the ethanolamine arms, plus the hydroxyl rotations — are each multi-well (gauche[±]/trans), so the joint target is combinatorially multimodal, which the wide-coverage pool must cover.

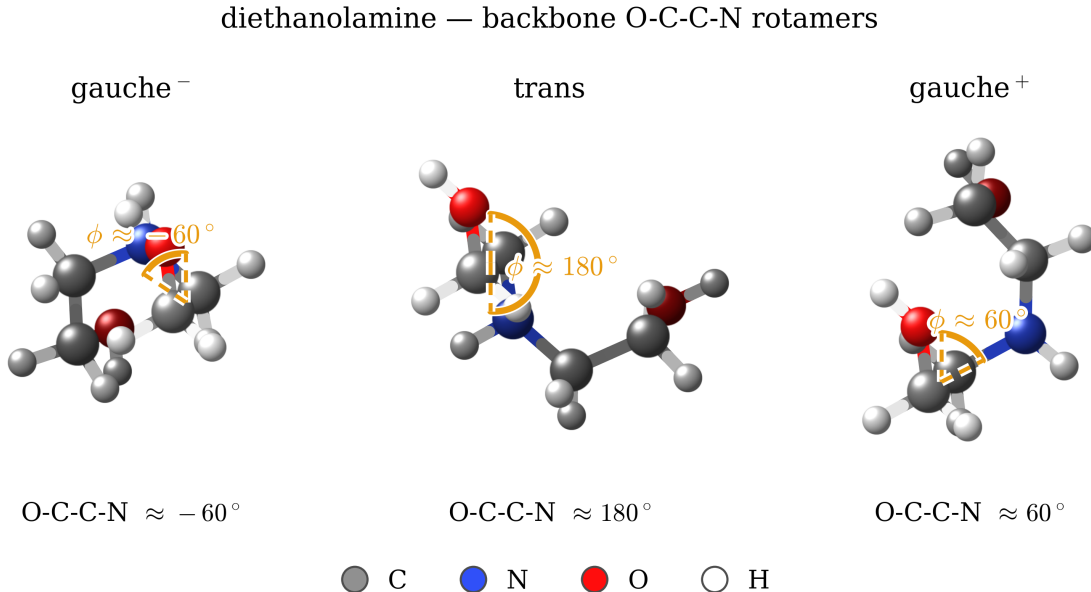


Figure 5: Typical diethanolamine configurations at the three states of a backbone O–C–C–N torsion: gauche[−] (−60°), trans (180°), gauche⁺ (+60°). The gold arc marks the dihedral ϕ ; ball-and-stick (C grey, N blue, O red, H white).

Against the MD reference (Fig. 6) the two losses are visually indistinguishable — both recover

diethanolamine ($d = 48$) — torsion marginals: BG vs annealed SMC reference

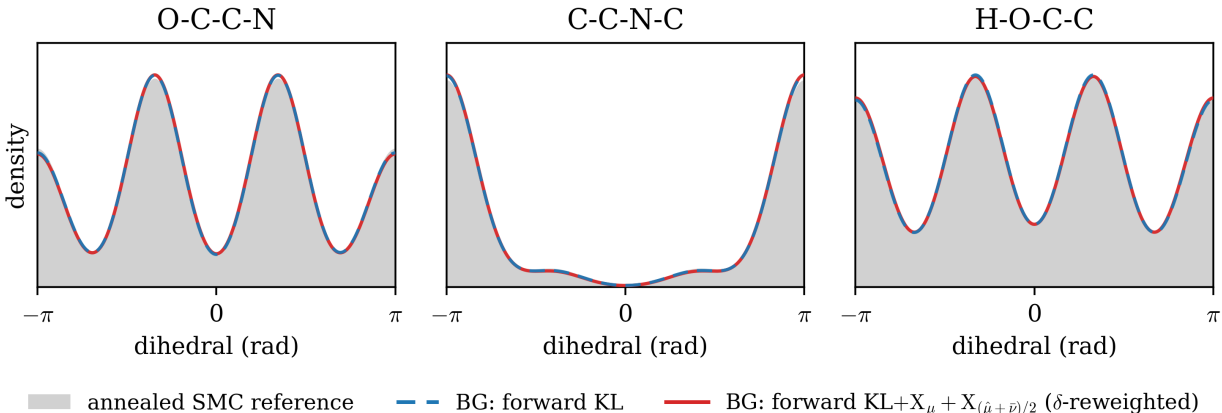


Figure 6: Diethanolamine torsion marginals of both generators against a long, converged molecular-dynamics reference (grey; a batched Langevin on the same target, 2×10^5 walkers): forward KL (blue) and $\text{KL} + X_\mu + X_{(\hat{\mu} + \hat{\nu})/2}$, δ -reweighted (red) — both sharpened, the cap annealed to $e = 200$. Both recover the multimodal backbone torsions and track the MD reference, evidence that they sample the target rather than collapsing onto a single well.

the multimodal backbone torsions and neither is discernibly closer to MD. The separation is in efficiency, not in the sampled marginals: $\text{KL} + X$ reaches $t = 1$ in fewer stages (5 vs 6) with a higher final-step ESS (0.74 vs 0.53), a $\sim 2.6\times$ smaller error-propagation factor ($F = 35.9$ vs 92.6), and fewer rejected stage attempts in the early ladder (stages 2–4: 1, 1, 0 rejections versus forward KL’s 2, 2, 1). Switching the trained flow off entirely — the identity map at every stage, i.e. plain adaptive annealed SMC over the same ladder — gives $F = 1455$ in $K = 10$ stages. The trained X -regularized flow therefore cuts the propagation factor by $\sim 40\times$ and halves the ladder length (5 vs 10): on this singular target the transport map adds even more over annealing alone than on glycerol, the practical justification for training a flow at all.

1.4.3 Alanine dipeptide ($d = 60$)

We push the generator to alanine dipeptide (ADP, Ac-Ala-NHMe), the canonical Boltzmann-generator benchmark, in full internal coordinates ($d = 60$). The qualitative step up from glycerol and diethanolamine is the target’s character: those are dominated by smooth rotatable-bond multimodality, whereas ADP folds the backbone tightly enough that the Lennard-Jones clash core — not the torsion profile — sets the hardest part of the anneal. We report the X -regularized loss only ($\text{KL} + X_\mu + X_{(\hat{\mu} + \hat{\nu})/2}$, δ -reweighted, $\delta = 0.1$), with a per-stage Monte-Carlo sharpening in which *both* the cap $e: 100 \rightarrow 200$ and the clash floor $r_{\text{floor}}: 0.2 \rightarrow 0.1$ are annealed arithmetically (the only target that anneals r_{floor}). The run reaches $t = 1$ in $K = 11$ stages at $F = 67.9$, with every stage holding validation ESS ≥ 0.51 and sharpening ESS ≥ 0.95 (Table 4, Fig. 7).

For comparison, the lighter targets close at $F = 14.2$ (glycerol) and $F = 35.9$ (diethanolamine) with the same loss; ADP’s larger F is the price of its singular nonbonded core, not of its multimodality. Three ADP-only factors nonetheless make this F read smaller than a strict like-for-like extension of that trend: the per-stage ESS is the smooth estimator of §1.3 rather than the raw weighted ESS (WSS); r_{floor} is annealed alongside the cap; and the ladder is much longer ($K = 11$

loss	K	wall (min)	F
annealed SMC	12	23	129.70
KL+ X_μ + $X_{(\hat{\mu}+\bar{\nu})/2}$, δ -rw.	11	~ 1000	67.95

Table 4: Alanine dipeptide ($d = 60$), the X-regularized loss with a per-stage Monte-Carlo sharpening reweight (the cap e and floor r_{floor} annealed arithmetically): ladder length K , wall-clock, and the error-propagation factor F (4) (smaller better; F includes the sharpening reweight). The first row is a no-flow annealed-SMC reference (identity map over the same ladder); the trained flow only halves F here ($130 \rightarrow 68$), far less than the $20\text{--}40\times$ reduction on the lighter $d \leq 48$ targets — it earns little near ADP’s clash singularity. The bare composed flow’s end-to-end ESS against the sharp target is $\sim 5 \times 10^{-6}$, so the per-stage SMC sharpening is the deliverable, not a raw pushforward. δ -rw. = δ -reweighted quench and temper.

versus 4 and 5), its smaller per-step jumps keeping each per-stage ESS high. All three raise the per-stage ESS entering F , and none applied to the lighter targets.

The $t \approx 0.3\text{--}0.45$ bottleneck. The instability is sharply localized. The stages spanning $t \approx 0.29\text{--}0.43$ (soft cap $e \approx 129\text{--}143$) are where the nonbonded core first bites: the run minimum validation ESS (0.51 at $t = 0.39$) sits here, and these are the *only* stages that shrink the step ($n_{\text{shrink}} = 1$ at stages 3, 4, 5) or retry heavily (3 attempts at stages 5, 6). Below $t \approx 0.3$ the torsions still dominate and the flow trains in a single attempt; above $t \approx 0.5$ the cap is sharp enough that the basins separate cleanly and ESS recovers to ≈ 0.8 . The binding constraint for ADP is the clash singularity, not the multimodality — the qualitative difference from the $d \leq 48$ molecules, and the reason the three devices of §1.3 (particle drop, best-ESS gate snapshots, r -floor anneal), inert on glycerol and diethanolamine, were necessary here.

A key observation (open difficulty). The per-stage ESS history (Fig. 7) exposes two coupled instabilities that the best-ESS gate works around but does not cure. First, the within-stage ESS is *non-monotone and unstable*: it can rise, plateau, then drop abruptly — the sharp collapse late in the $t = 0.65$ stage and the volatility at $t = 1$ are the clearest cases — so the last training step is frequently far below the best (stage 9’s training ESS decays to 0.03 by step 500 while the best-ESS gate keeps a snapshot at validation ESS = 0.80). Second, near the clash bottleneck ($t \approx 0.3\text{--}0.5$) several flows *degenerate toward the identity map*: the best achievable ESS barely exceeds the $t = 0$ no-flow baseline ($\Delta = \text{best} - \text{identity} \approx 0$ at stages 3, 4, 7), i.e. training fails to improve on doing nothing. Both are understood consequences of the *singularity of the potential* — the Lennard-Jones $r \rightarrow 0$ divergence leaves the soft-capped target ill-conditioned exactly where the nonbonded core first bites, so the gradient signal that would pull the flow off identity is weakest there. We report this as a known limitation; robust remedies (stronger or adaptive regularization annealing, parameterizations tailored to the repulsive wall) are left to future work.

Multimodality and the L/D mirror. ADP’s backbone ϕ, ψ are multi-basin (Fig. 8). A second, ADP-specific feature is chirality: the classical force field is mirror-symmetric (every term in (1) is invariant under reflection), so the L- and D-forms are isoenergetic. A physical MD trajectory stays in one form — inverting the C_α needs bond breaking — but the internal-coordinate flow carries no chirality constraint and emits *both* (Fig. 9). In the Ramachandran (Fig. 10) this appears as the $(\phi, \psi) \rightarrow (-\phi, -\psi)$ point symmetry: the D-basins are the exact mirror of the L-basins, each populated, where a single kinetically trapped physical trajectory would show only one.

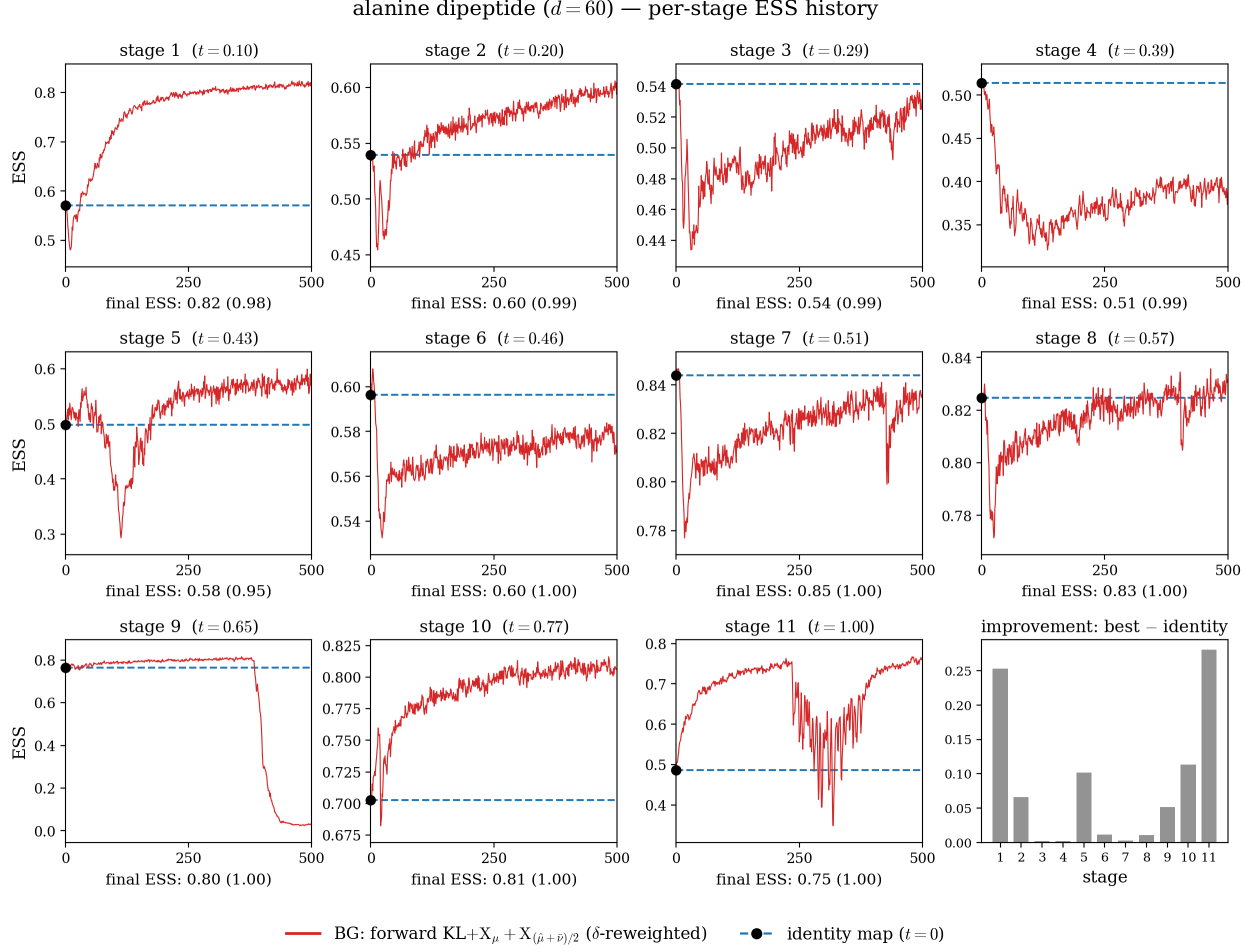


Figure 7: Alanine dipeptide per-stage ESS history: each panel is the per-step effective sample size during that stage’s X-regularized δ -reweighted flow training (red), against the identity-map baseline at $t = 0$ (blue dashed). Below each panel is that stage’s accepted validation ESS with the sharpening ESS in parentheses. Note the dip across $t \approx 0.29$ – 0.46 — the clash bottleneck — and the recovery to $\text{ESS} \approx 0.8$ once the cap is sharp enough to separate the basins.

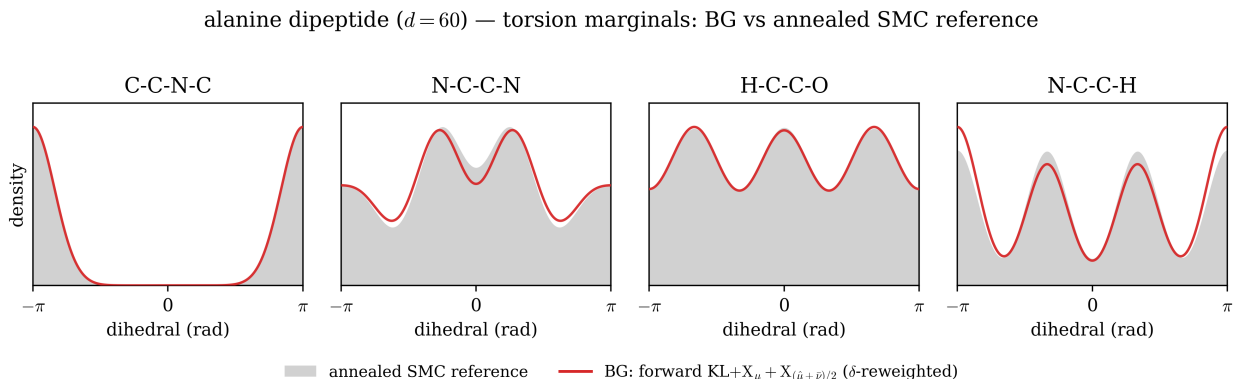


Figure 8: Alanine dipeptide torsion marginals of the BG deliverable (red) against a long, converged molecular-dynamics reference (grey; batched Langevin on the same sharp target, 2×10^5 walkers). The four most multimodal proper torsions are shown; the densities are $\pm\phi$ -symmetrised (the achiral target has $p(\phi) = p(-\phi)$). The BG recovers the multimodal backbone torsions and tracks MD; the peptide ω (C–C–N–C) is trans-dominant in the generator, the one torsion whose high rotation barrier also makes the uniform-init MD reference slow to converge.

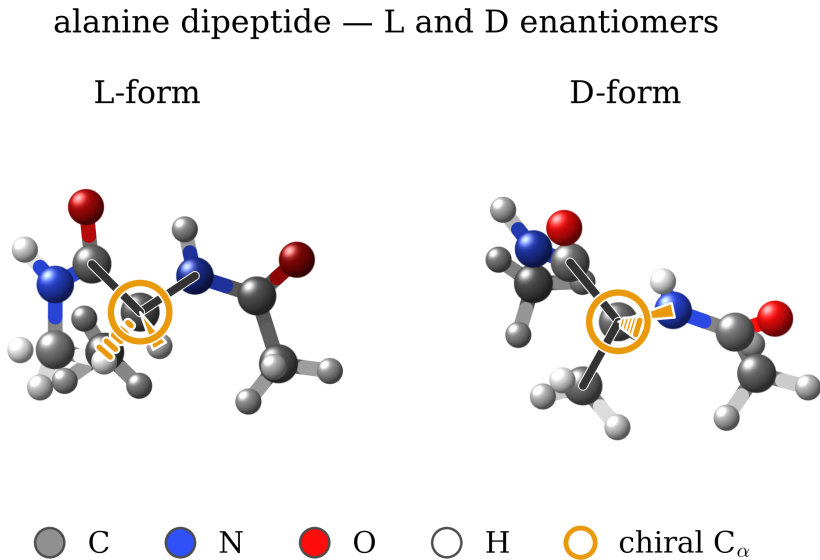
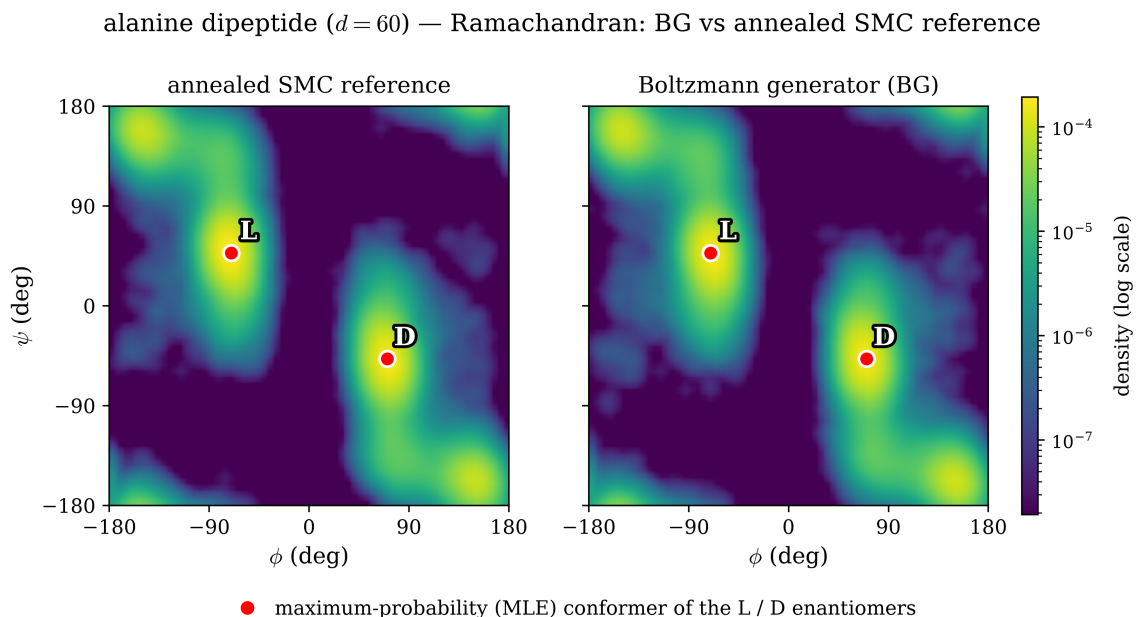


Figure 9: The two enantiomers of alanine dipeptide: the energy-minimized L-form and the D-form (a 3D reflection of it), shown from different viewpoints (ball-and-stick; C grey, N blue, O red, H white). The chiral C_α is ringed in gold with stereo bonds (solid wedge toward the viewer, hashed behind), making the opposite handedness explicit in each panel. The achiral Amber force field makes the two isoenergetic, so the internal-coordinate generator emits both — the structural companion to the $(\phi, \psi) \rightarrow (-\phi, -\psi)$ Ramachandran symmetry of Fig. 10.



BG: forward $\text{KL} + X_\mu + X_{(\hat{\mu} + \nu)/2}$ (δ -reweighted)

Figure 10: Alanine dipeptide Ramachandran (ϕ - ψ), shared log-density scale: the BG deliverable (right) against the annealed-SMC reference (left). Red dots mark the maximum-probability (MLE) conformer of each enantiomer (L at $\phi < 0, \psi > 0$; D at $\phi > 0, \psi < 0$). Both panels populate the same backbone basins with the $(\phi, \psi) \rightarrow (-\phi, -\psi)$ point symmetry: the achiral (mirror-symmetric) vacuum force field makes the L- and D-enantiomers isoenergetic, and the internal-coordinate generator, carrying no chirality constraint, samples both (§1.1).

Conclusion. The X-regularized BG closes the full internal-coordinate ADP target to $t = 1$ in 11 stages at $F = 67.9$ on a consumer GPU, recovering the multimodal backbone torsions against a converged MD reference and emitting both chiral forms of the mirror-symmetric target. The lesson carried over from the lighter $d \leq 48$ molecules is inverted: ADP’s difficulty is not its combinatorial torsion multimodality but its nonbonded clash core, concentrated in the $t \approx 0.3\text{--}0.45$ stages, which the particle drop, best-ESS gate snapshots, and r -floor anneal together make tractable. Across all three molecules the pattern is consistent: the sharpening split makes the singular target trainable at all, and the X-regularized flow then earns a 30–40 $\times$ reduction in importance-weight waste where the landscape is multimodal but clash-mild, narrowing to $\sim 2\times$ where the singularity itself dominates.

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